Bond Duration and Bank Equity

LECTURE 12 · REVIEW NOTES
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A borrower can be safe and its bonds can still lose market value. Today we separate credit risk from interest-rate risk, measure the timing of cash flows, and then use duration to see why a small loss on a bank's assets can produce a large loss on its equity.

1. Connect the credit spread to the discount rate

$$y_c = y_g + s, \qquad s \approx PD(1 - R)$$

In our simplified example, annual PD is 2%, recovery is 40% and the government yield is 4%. The spread is 0.02 × 0.60 = 1.20%, or 120 bp, and the required yield is 5.20%. Other risk and liquidity premiums are set to zero. The historical chart adds rating OAS to a common 10-year Treasury yield as a classroom approximation; these are not the indices' reported YTMs.

2. The same cash flows are worth less at a higher yield

$$P = \sum_{j=1}^{2T} \frac{C}{(1 + y/2)^j} + \frac{FV}{(1 + y/2)^{2T}}$$

Here C is the semiannual coupon, y is annual nominal YTM, and T is maturity in years. Price is above par when YTM is below the coupon rate, at par when equal, and below par when YTM exceeds coupon. In Excel, use the rate per payment period and the number of payments.

For FV $1,000, annual coupon 6%, five years and YTM 5%: `=-PV(5%/2,5*2,1000*6%/2,1000,0)`. This gives $1,043.7603. The minus sign reports a positive purchase price; type 0 means payments arrive at period-end.

3. Macaulay duration measures timing

$$D_{Mac} = \sum_j t_j w_j, \qquad w_j = \frac{PV(CF_j)}{P}, \qquad t_j = j/2$$

First discount every payment. Add the present values to get price. Divide each present value by that price, then multiply the weight by its time in years. The weights sum to one. A zero-coupon bond has duration equal to maturity; a coupon bond delivers some value earlier.

Do not call this a guaranteed payback date. Macaulay is the average payment time using present-value weights. It does not tell us a guaranteed sale price or the date cumulative cash receipts recover the original purchase price.

From cash-flow timing to price sensitivity

4. The bond worksheet and portfolio example

In the five-year worksheet, the 6% coupon gives $30 every half-year and $1,030 at maturity. At 5% YTM, price is $1,043.7603 and Macaulay duration is 4.4084 years. At the same yield and maturity, increasing the coupon shifts more value to earlier payments and lowers duration; reducing it shifts weight toward maturity.

For a portfolio, use current market-value weights. With $200,000 at duration 1.5 years, $300,000 at 4 years and $500,000 at 8 years, the weights are 20%, 30% and 50%.

$$D_{Mac,P} = 0.20 \times 1.5 + 0.30 \times 4 + 0.50 \times 8 = 5.50\ years$$

5. Modified duration measures price sensitivity

$$D_{mod} = \frac{D_{Mac}}{1 + y/2}, \qquad \Delta P \approx -D_{mod} \times P \times \Delta y$$

For the worksheet bond, Dmod = 4.4084075905 / 1.025 = 4.3008854541. A +1 percentage-point yield move means Δy = 0.01, so the approximate percentage price change is -4.3009%. Multiply that percentage change by current market value to obtain the dollar change. If modified duration is already given, do not divide by 1 + y/2 again.

Duration and beta both describe exposure to a risk factor. Duration links a yield move to a local price change; beta links equity to market returns, and CAPM links beta to expected return. They are different measures. A higher yield lowers a positive-duration bond’s price when cash flows are held fixed.

6. Why the bank example matters

SVB motivates the problem: long-lived assets can lose value when yields rise, and deposit withdrawals can require liquidity before those assets mature. The lecture’s bank calculation uses SVB-sized balance-sheet numbers, but the durations and common yield are hypothetical assumptions. It is not a reconstruction of SVB’s actual loss.

We assume fixed cash flows and a common parallel yield shock. Duration is a first-order approximation. Convexity, changing credit spreads and depositor behavior can make actual changes different.

Worked bank example: assets and liabilities

Use the same numbers throughout. Dollar values are in **billions**. Annual nominal yield is 5%, with semiannual compounding. Yields then rise to 6%.

Position	Market value ($bn)	Macaulay duration
Assets	211.8	3.00 years
Liabilities	195.8	0.50 years

Step 1. Calculate equity before doing duration

$$E = A - L = 211.8 - 195.8 = 16.0$$

Equity is the residual claim. It is not another asset to add to the balance sheet. The bank starts with $16bn available to absorb a net decline in asset value.

Step 2. Convert each side to modified duration

$$D_{mod,A} = \frac{3}{1 + 0.05/2} = 2.9268292683$$

$$D_{mod,L} = \frac{0.5}{1 + 0.05/2} = 0.4878048780$$

Both denominators are 1.025. Keep full precision; rounding durations too early makes the later equity calculation drift.

Step 3. Apply the same yield shock to each side

$$\Delta y = 0.06 - 0.05 = 0.01$$

$$\Delta A \approx -2.9268292683 \times 211.8 \times 0.01 = -6.1990243902$$

$$\Delta L \approx -0.4878048780 \times 195.8 \times 0.01 = -0.9551219512$$

Assets fall by about $6.1990bn; liabilities fall by about $0.9551bn. New assets are $205.6010bn and new liabilities are $194.8449bn. The liability decline offsets part of the asset loss when we calculate equity. These are economic market values, not automatic changes in book accounting values.

The central result: equity modified duration

Step 4. Subtract liability dollar duration from asset dollar duration

$$D_{mod,E} = \frac{A \times D_{mod,A} - L \times D_{mod,L}}{E}$$

$$D_{mod,E} = \frac{211.8 \times 2.9268292683 - 195.8 \times 0.4878048780}{16} = 32.7743902439$$

The multiplications matter: we weight each duration by its position's market value. Liabilities enter with a **minus sign**, because equity equals assets minus liabilities. Dividing by the relatively small equity value magnifies the sensitivity.

$$D_{mod,E} = \frac{A}{E} D_{mod,A} - \frac{L}{E} D_{mod,L}$$

This is not an ordinary average of the two durations. The asset weight exceeds one and the liability weight is negative. Equity duration can therefore exceed both asset and liability duration. It is also **not a 32.77-year payback horizon**.

Step 5. Calculate the equity change in two equivalent ways

$$\Delta E = \Delta A - \Delta L = -6.1990243902 - (-0.9551219512) = -5.2439024390$$

$$\Delta E \approx -D_{mod,E} \times E \times \Delta y = -32.7743902439 \times 16 \times 0.01$$

Both routes give a loss of $5.2439bn. Subtracting a negative liability change is essential: do not add the two losses.

$$\frac{\Delta E}{E} \approx -32.7743902439 \times 0.01 = -32.7744\%$$

$$E_{new} = 16 - 5.2439024390 = 10.7560975610$$

Final check: new assets minus new liabilities gives the same $10.7561bn of equity. Assets fall only about 2.93%, but equity falls about 32.77% because leverage leaves a thin equity cushion.

Before submitting: distinguish Macaulay from modified duration; write the yield shock as a decimal; keep all values in the same dollar units; subtract liabilities; and divide the percentage equity loss by initial equity, not total assets.